Chapter 18, Problem 25P

Problem

Obtain the inverse Fourier transform of the following signals.

(a) $G(\omega) = \dfrac{5}{j\omega - 2}$

(b) $H(\omega) = \dfrac{12}{\omega^2 + 4}$

(c) $X(\omega) = \dfrac{10}{(j\omega - 1)(j\omega - 2)}$

Step-by-step solution

Step 1 of 3

(a)

Consider the following Fourier transform expression.

$$G(\omega) = \frac{5}{j\omega - 2}$$

Apply inverse Fourier transform to this expression.

$$\mathcal{F}^{-1}\left[G(\omega)\right] = \mathcal{F}^{-1}\left[\frac{5}{j\omega - 2}\right]$$

$$\begin{aligned} g(t) &= 5\mathcal{F}^{-1}\left[\frac{1}{j\omega - 2}\right] \\ &= 5\mathcal{F}^{-1}\left[\frac{1}{(-2) + j\omega}\right] \\ &= 5e^{-(-2)t}u(t) \qquad \left[\text{Since, } \mathcal{F}\left[e^{-at}u(t)\right] = \frac{1}{(a + j\omega)}\right] \\ &= 5e^{2t}u(t) \end{aligned}$$

$$g(t) = 5e^{2t}u(t)$$

Therefore, the required inverse Fourier transform of the function is $\boxed{5e^{2t}u(t)}$.

Step 2 of 3

(b)

Consider the following Fourier transform expression.

$$H(\omega)=\frac{12}{\omega^2+4}$$

Apply inverse Fourier transform to this expression.

$$\mathcal{F}^{-1}[H(\omega)]=\mathcal{F}^{-1}\left[\frac{12}{\omega^2+4}\right]$$

$$\begin{aligned}
h(t)&=3\mathcal{F}^{-1}\left[\frac{4}{\omega^2+2^2}\right]\\
&=3\mathcal{F}^{-1}\left[\frac{2(2)}{\omega^2+2^2}\right]\\
&=3\left[e^{-2|t|}u(t)\right] \qquad \left[\text{Since, } \mathcal{F}\left[e^{-a|t|}u(t)\right]=\frac{2a}{(a^2+\omega^2)}\right]\\
&=3e^{-2|t|}u(t)
\end{aligned}$$

Therefore, the required inverse Fourier transform of the function is $\boxed{3e^{-2|t|}u(t)}$.

Step 3 of 3

(c)

Consider the following Fourier transform expression.

$$X(\omega)=\frac{10}{(j\omega-2)(j\omega-1)}$$

Apply inverse Fourier transform to this expression.

$$\mathcal{F}^{-1}[X(\omega)]=\mathcal{F}^{-1}\left[\frac{10}{(j\omega-2)(j\omega-1)}\right]$$

$$\begin{aligned}
x(t)&=\mathcal{F}^{-1}\left[\frac{10}{(j\omega-2)}-\frac{10}{(j\omega-1)}\right]\\
&=\mathcal{F}^{-1}\left[\frac{10}{(j\omega-2)}\right]-\mathcal{F}^{-1}\left[\frac{10}{(j\omega-1)}\right]\\
&=10\mathcal{F}^{-1}\left[\frac{1}{(-2)+j\omega}\right]-10\mathcal{F}^{-1}\left[\frac{1}{(-1)+j\omega}\right]\\
&=10\left[e^{-(-2)t}u(t)\right]-10\left[\left[e^{-(-1)t}u(t)\right]\right] \qquad \left[\text{Since, } \mathcal{F}\left[e^{-at}u(t)\right]=\frac{1}{(a+j\omega)}\right]\\
&=10\left\{e^{2t}-e^{t}\right\}u(t)
\end{aligned}$$

Therefore, the required inverse Fourier transform of the function is $\boxed{10\left\{e^{2t}-e^{t}\right\}u(t)}$.